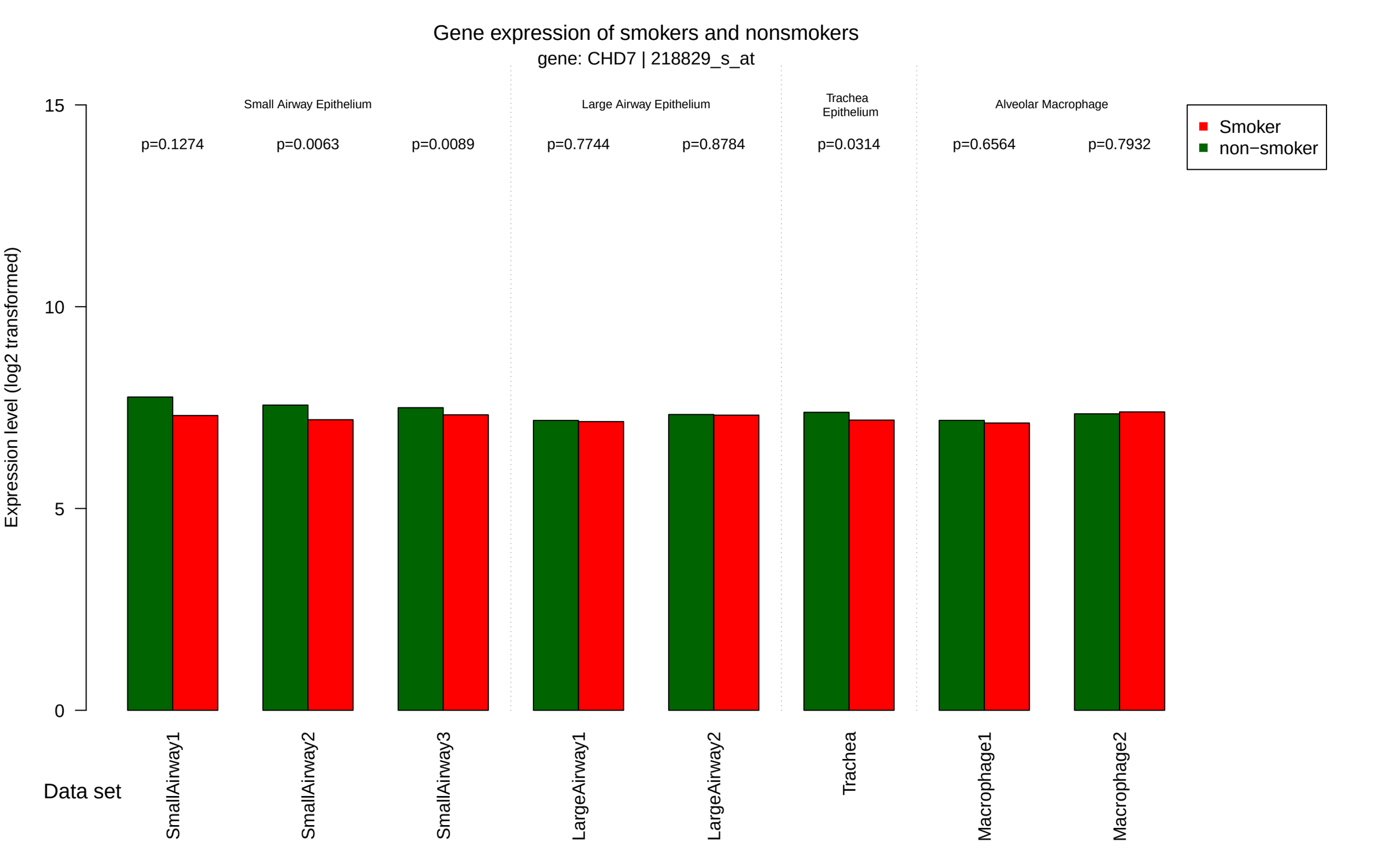
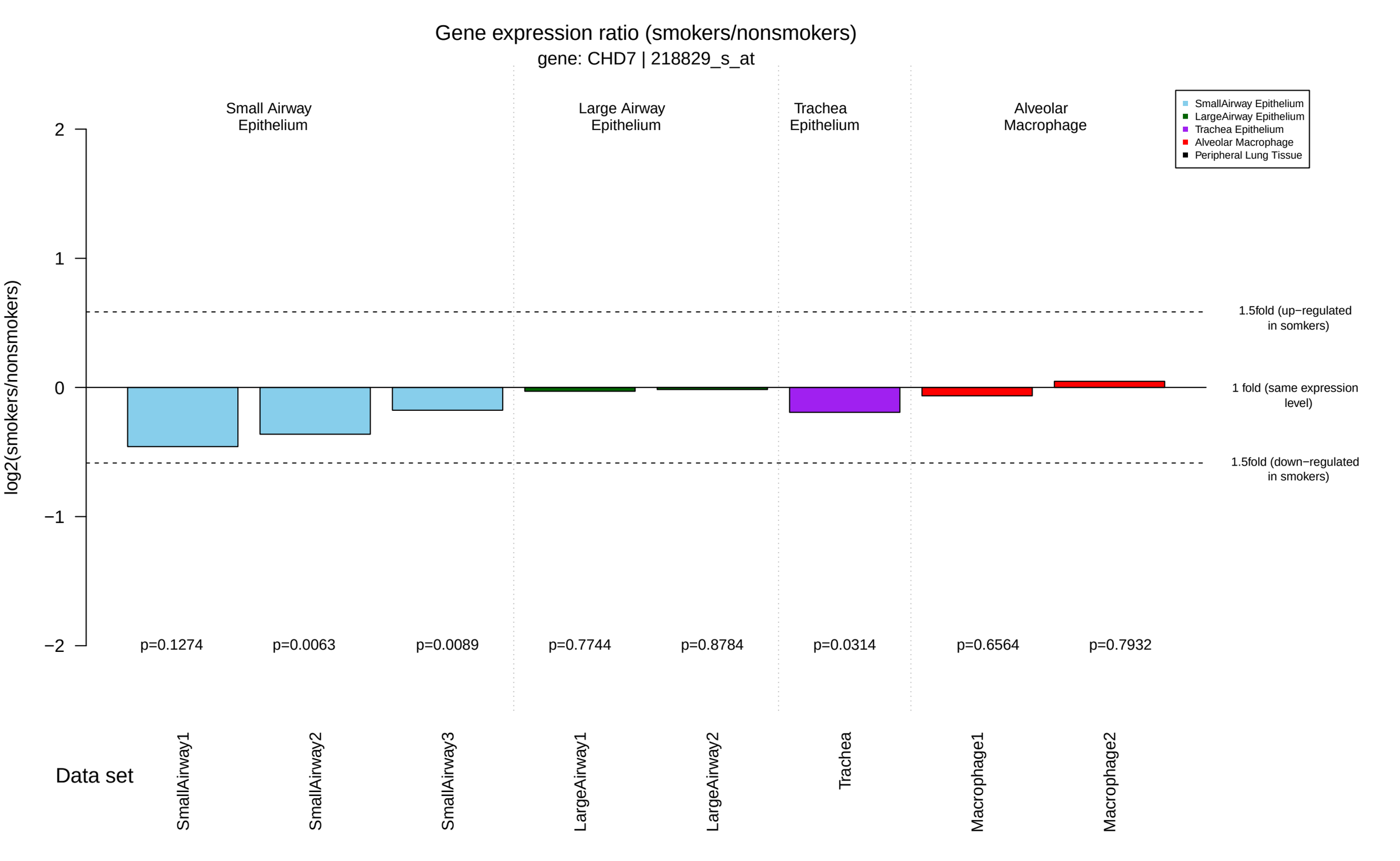
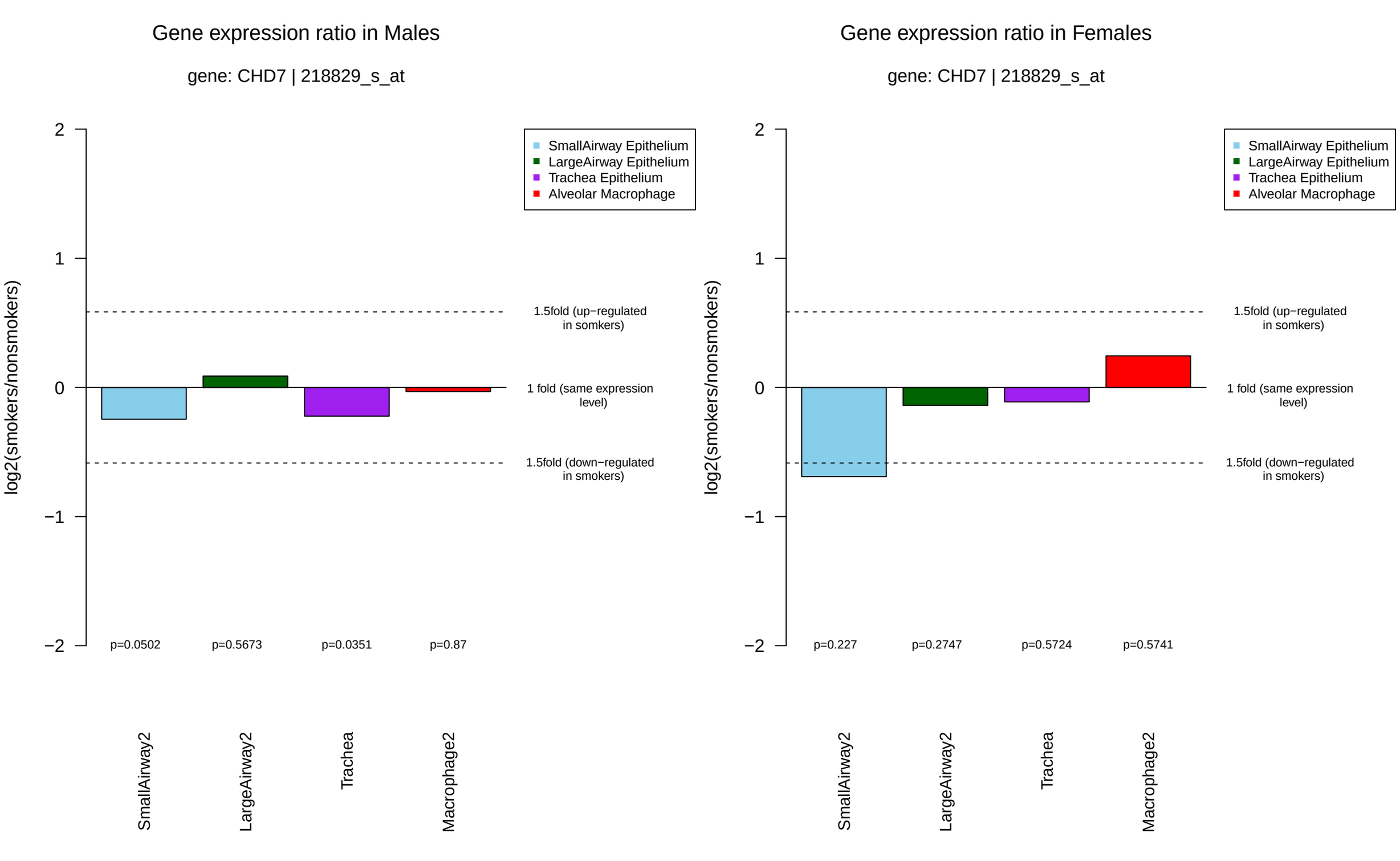
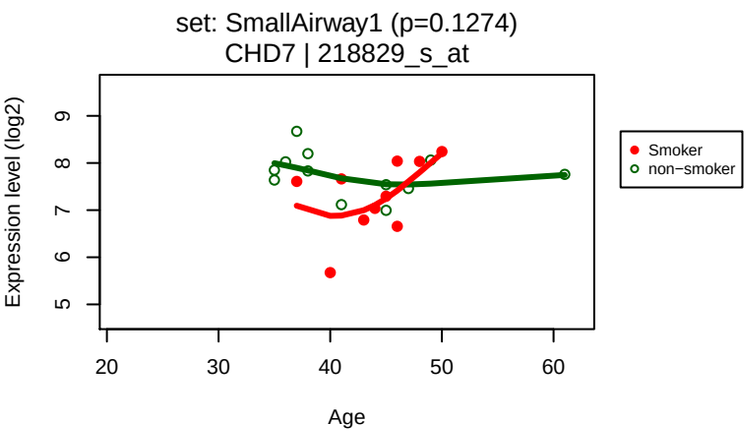
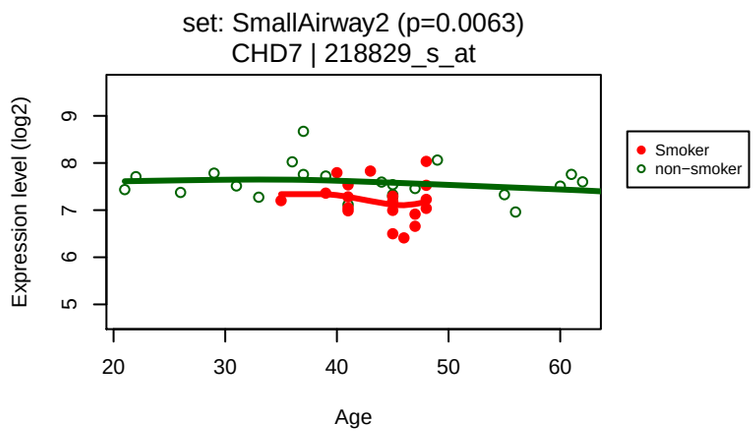
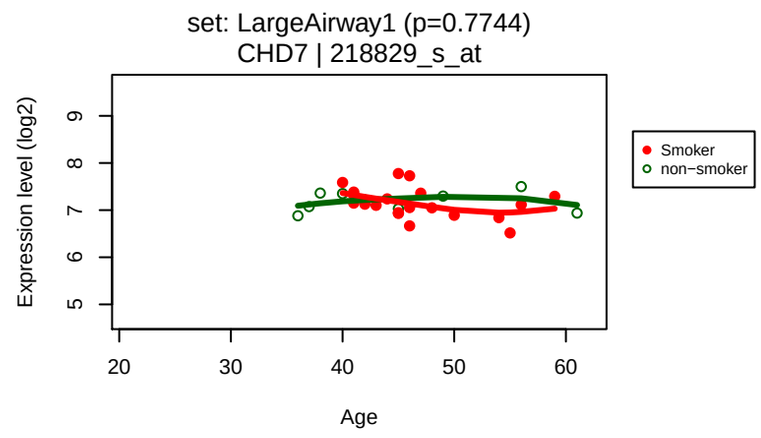

Gene expression of smokers and nonsmokers
gene: CHD7 | 218829_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1274
p=0.0063
p=0.0089
p=0.7744
p=0.8784
p=0.0314
p=0.6564
p=0.7932
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

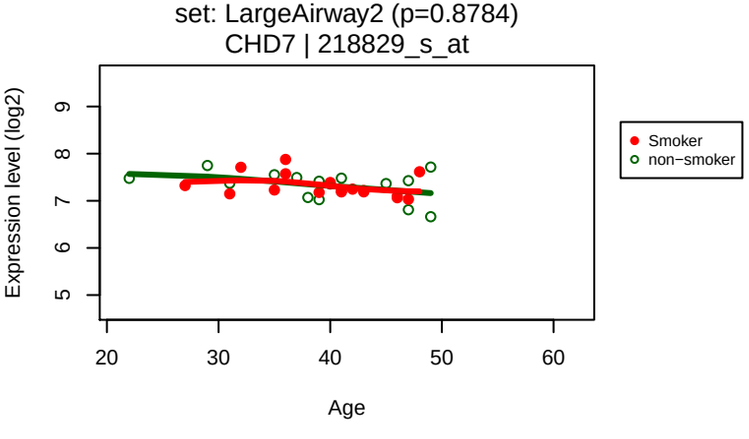

Gene expression ratio (smokers/nonsmokers)
gene: CHD7 | 218829_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1274
p=0.0063
p=0.0089
p=0.7744
p=0.8784
p=0.0314
p=0.6564
p=0.7932
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

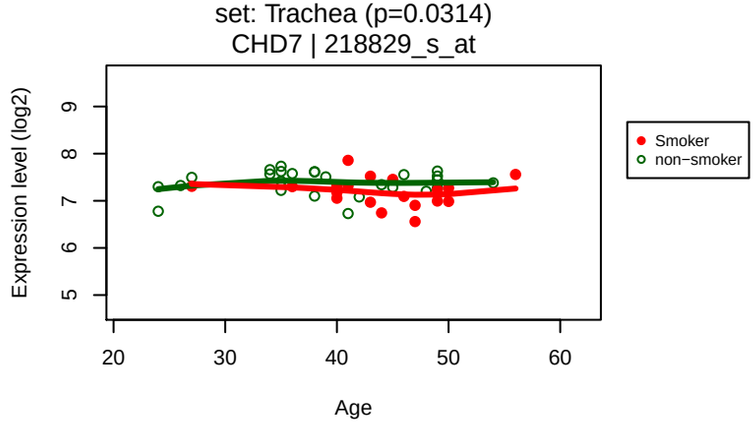

Gene expression ratio in Males
gene: CHD7 | 218829_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0502
p=0.5673
p=0.0351
p=0.87
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHD7 | 218829_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.227
p=0.2747
p=0.5724
p=0.5741
SmallAirway2
LargeAirway2
Trachea
Macrophage2

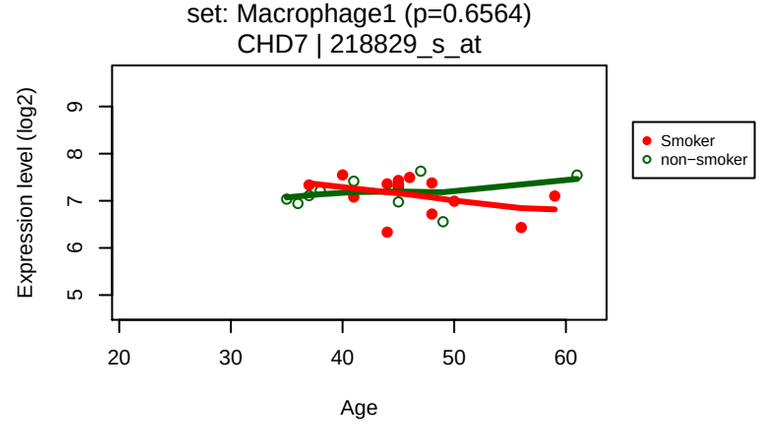

set: SmallAirway1 (p=0.1274)
CHD7 | 218829_s_at
Expression level (log2)
Age
Smoker
non−smoker

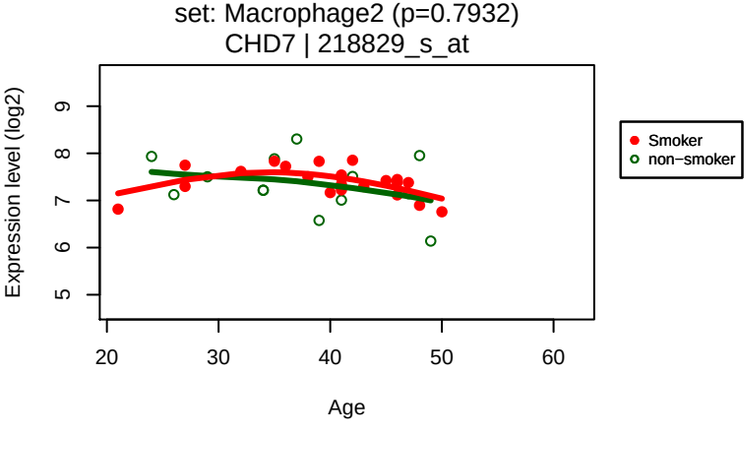

set: SmallAirway2 (p=0.0063)
CHD7 | 218829_s_at
Expression level (log2)
Age
Smoker
non−smoker

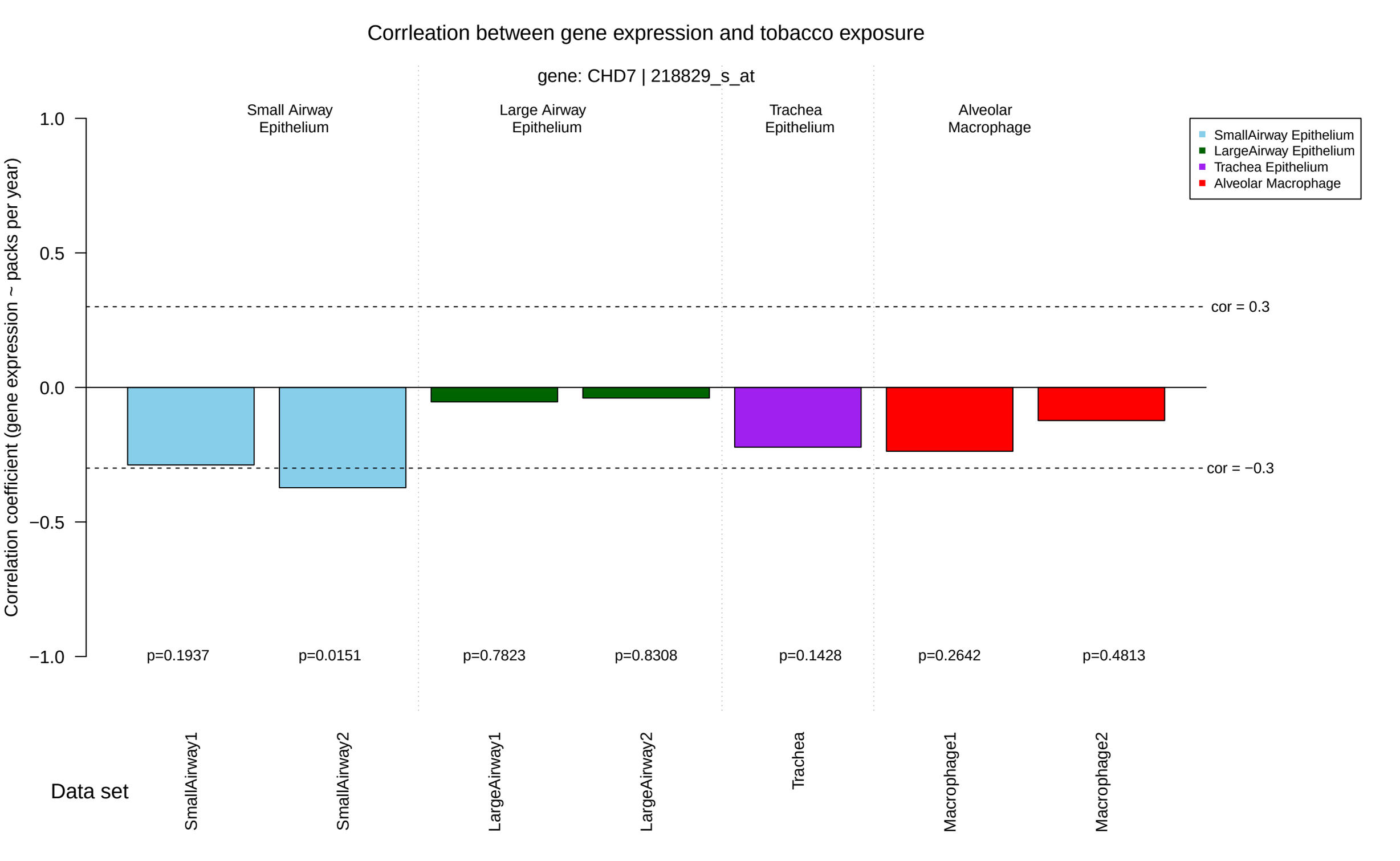

set: LargeAirway1 (p=0.7744)
CHD7 | 218829_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8784)
CHD7 | 218829_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0314)
CHD7 | 218829_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6564)
CHD7 | 218829_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7932)
CHD7 | 218829_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHD7 | 218829_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1937
p=0.0151
p=0.7823
p=0.8308
p=0.1428
p=0.2642
p=0.4813
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2